

Humans Do Not Delve: Measuring LLM-Induced Style Drift in 25 Years of Speech Conference Papers

Ryota Nishimura

Toyohashi University of Technology, Japan
nishimura.ryota.tz@tut.jp

Abstract

Large language models (LLMs) are now widely used to draft and polish scientific text, and readers have started to recognize a characteristic “AI style.” In this paper, we measure that style against 25 years of ground truth. We analyze 62,977 papers (166.9 million words) from the complete ICASSP 2001–2026 and Interspeech 2007–2025 proceedings, and we separate a clean human baseline (2015–2022) from the LLM era (2024–2026). The baseline is remarkably stable: for 24 years, em-dash frequency, semicolon rate, and sentence length barely moved. In 2024 the picture changes abruptly. Words such as *underscoring* rise by two orders of magnitude, *notably* rises sevenfold, and em-dashes triple in 2025 in both conferences. We also observe that infamous markers such as *delve* peaked in 2024 and then declined, so fixed blocklists decay quickly. On the other hand, phrases that humans use routinely, such as *in other words*, are vanishing from the contaminated corpus and turn out to be strong human markers. We distill these measurements into evidence-based writing guidelines, encode them as a reference file for coding agents, and show in a controlled generation experiment that the guidelines move AI-drafted introductions to within the measured human range. This paper itself was written under those guidelines.

Index Terms: scientific writing, large language models, style analysis, diachronic corpus study, AI-assisted writing

1. Introduction

Writing assistance by large language models has moved from novelty to routine in only three years. Surveys of recent scholarly text estimate that a substantial fraction of new abstracts and reviews contain LLM-generated or LLM-edited passages [1, 2]. Readers, meanwhile, have developed an informal detector of their own. Certain words, punctuation habits, and paragraph rhythms are now read as “AI style,” sometimes unfairly, and authors who write naturally in that register risk being suspected of outsourcing their prose.

Most quantitative evidence for this drift comes from biomedical abstracts and general scholarly corpora [3, 2, 4]. To the best of our knowledge, no comparable study exists for the speech and language processing community, even though this community both builds LLMs and writes with them. A field-specific study matters for two reasons. First, editorial norms differ across fields, so a marker that is rare in medicine may be normal in engineering, and vice versa. Second, practical guidance for AI-assisted writing needs field-specific target values, not just a list of forbidden words.

In this paper, we present a diachronic analysis of the complete ICASSP and Interspeech proceedings, 62,977 papers and 166.9 million words in total, spanning 2001 to 2026. The corpus straddles the release of ChatGPT in late 2022, which lets us treat the years up to 2022 as a clean human baseline and the years from 2024 onward as a naturally contaminated condition. We ask three questions. We first ask how stable the writing conventions of the field

were before LLMs arrived. We then ask which lexical and typographic markers changed after 2023, and how those markers themselves evolve over time. Note that the second question has a moving target: as certain words become notorious, both model vendors and authors suppress them, so any fixed blocklist should decay. Finally, we ask whether the resulting measurements can steer an LLM back into the human distribution.

The contributions are threefold. We provide the first large-scale, field-specific measurement of LLM-induced style drift in speech conference writing, including evidence that the drift is visible in punctuation alone. We show that AI-marker vocabulary is subject to fashion dynamics, with a distinct peak-and-decline pattern for the most notorious words, and we identify declining human phrases as a more durable signal. In order to make the findings actionable, we convert them into quantitative writing guidelines, release them as a machine-readable style file for LLM coding agents, and validate them in a generation experiment. The rest of the paper is organized as follows. Section 2 describes the corpus and the measurement pipeline. Section 3 reports the diachronic results. Section 4 derives the guidelines, and Section 5 evaluates them. Section 6 concludes.

2. Corpus and method

2.1. Corpus

We collected the complete proceedings of ICASSP 2001–2026 (45,571 papers, 114.0M words) and Interspeech 2007–2025 (17,406 papers, 52.9M words). Laboratory archives of the distributed proceedings media provided most years, and

the ISCA Archive provided the years that were published online only. Text was extracted with `pdftotext`. For each paper we kept the body from the abstract heading to the references heading, repaired end-of-line hyphenation, and discarded files shorter than 2,000 characters. Files that survived this filter are counted as papers above.

2.2. Three-layer design

A naive approach would pool all years and describe “how the field writes.” This is no longer safe, because papers from 2023 onward already contain LLM-assisted prose. In other words, recent human-authored proceedings cannot serve as a style reference. We therefore split the corpus into three layers. The **core** layer, 2015–2022 (19,925 papers, 56.8M words), predates ChatGPT and defines the human baseline used for all target frequencies. The **contaminated** layer, 2024–2026, is not used as a reference at all. Instead it acts as a detector, because any expression that surges there while the baseline was flat is a candidate AI marker. The ICASSP 2023 edition, submitted in late 2022, sits on the boundary and is excluded from both layers. The **historical** layer, 2001–2014, tells us whether a baseline convention is a recent fashion or a stable property of the field.

2.3. Measurements

For every conference and year we computed word and n -gram frequencies, punctuation rates, sentence-length statistics, and section-heading sequences, all normalized per million words (/M). Candidate AI markers were taken from prior reports [3, 4] and from our own comparison of LLM-drafted and human introductions (Section 5). For each candidate we report the core-layer baseline and the 2024–2026 mean, and we classify it by two thresholds: expressions with a baseline below 3/M are labeled **BAN** (humans essentially never write them), and expressions with a healthy baseline but a post-LLM surge factor of at least 3 are labeled **LIMIT** (humans use them, at a rate that should not be exceeded).

3. Results

3.1. The human baseline is remarkably stable

Table 1 summarizes conventions that stayed essentially constant from 2001 to 2024. Mean sentence length varied only between 23.2 and 25.4 words over 26 ICASSP editions. Semicolons appear about once every 30 sentences. Em-dashes appear 92–149 times per million words, which is roughly 0.4 per paper, and rhetorical questions are almost absent. It can be seen that these are not arbitrary stylistic choices but long-standing norms of the community, and this stability is what makes the post-2023 discontinuity measurable.

3.2. The formulaic core of the field

Beyond punctuation, the baseline contains a stable inventory of formulaic expressions that signal membership in the community. Table 2 lists representative examples with their core-layer rates in each venue. Some of them, such as *in this paper, we* and *the rest of the paper is organized*

Table 1: Stable properties of the human baseline (core layer; historical layer confirms stability back to 2001).

Metric	Human baseline
Sentence length (mean / median, words)	24.5 / 21
Semicolons per sentence	0.03–0.05
Em-dashes per million words	92–149
Em-dashes per paper	≈ 0.4
Bullet characters per paper	< 1
Question marks per 1k words	0.23

Table 2: Field formulas in the core layer (per million words).

Expression	ICASSP	Interspeech
in this paper(,) we	493	437
in order to	442	>300
state-of-the-art	360	358
in this work(,) we	205	196
in this section(,) we	180	81
on the other hand	154	135
as shown in Fig.~/Figure~	153	86
(the) paper is organized as (follows)	66	52
to the best of our knowledge	49	50

as follows, are openly formulaic, and style guides outside engineering often discourage them. In this field they are simply how introductions are written, and they have been for the whole historical layer. Others are quieter. The connective *in order to* appears 442 times per million words in ICASSP, although a general-purpose editor would shorten it to *to*. It can be seen from Section 5 that an unguided LLM reproduces the loud formulas but drops the quiet ones, which is one reason its drafts feel subtly foreign.

3.3. The 2024 discontinuity

The first LLM-era submission cycle produced an abrupt break. Table 3 shows representative trajectories. The adverb *notably* had drifted slowly from 6/M (2001) to 24/M (2020). In ICASSP 2024 it jumped to 152/M, and it reached 170/M in 2026. The gerund *underscoring* rose from under 1/M to 50/M, a factor above 100. The adjective *crucial* had held near 40/M for fifteen years and tripled within two. Punctuation moved as well: em-dash frequency, flat for 24 years, more than tripled in 2025 in both conferences at once (514/M in ICASSP, 489/M in Interspeech), and bullet characters doubled. On the other hand, the average sentence length barely moved, so surface statistics alone do not capture everything that changed.

Figure 1 plots two of these trajectories over the full corpus. The visual impression matches the numbers: both curves are essentially flat for two decades, and both leave the historical band within one or two submission cycles of the ChatGPT release. Interspeech replicates every ICASSP trend with a one-year-aligned profile, which rules out venue-specific template artifacts as an explanation. Note that our estimates are conservative lower bounds on prevalence,

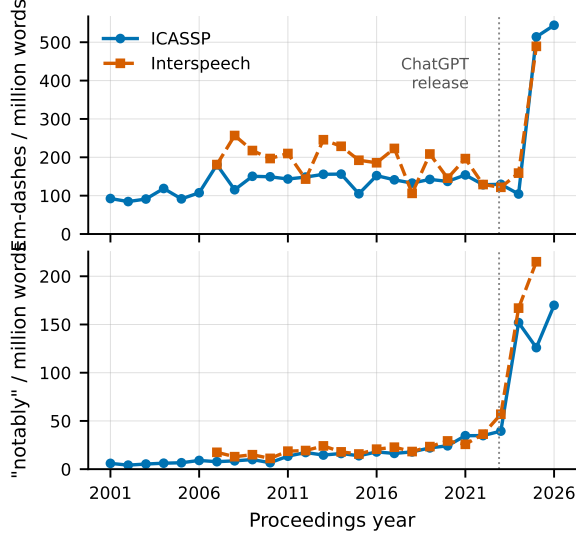


Figure 1: Em-dash rate (top) and frequency of *notably* (bottom) per million words in ICASSP (2001–2026) and Interspeech (2007–2025) proceedings. The dotted line marks the release of ChatGPT. Both metrics stay within their historical band for two decades and depart from it within one or two submission cycles.

Table 3: Per-million-word frequencies in ICASSP by year. IS25 gives the Interspeech 2025 value for comparison.

Expression	2020	2023	2024	2025	2026	IS25
em-dash (—)	137	130	104	514	544	489
<i>notably</i>	24	40	152	126	170	215
<i>crucial</i>	62	80	193	218	200	236
<i>underscoring</i>	0.4	0.2	13	22	50	—
<i>intricate</i>	4.1	4.9	52	40	22	—
<i>pivotal</i>	2.4	4.2	41	22	15	—
<i>delve</i>	0.4	1.3	12	6.2	1.2	—
<i>in the realm of</i>	0.2	0.8	15	6.7	1.2	—

since we measure marker frequency rather than the fraction of affected papers.

3.4. Marker fashion: peak and decline

The lower half of Table 3 shows a second-order effect. The most publicized AI words, *delve*, *realm*, *pivotal*, and *intricate*, peaked in 2024 and have been falling since, presumably because model vendors and authors now suppress them. Less notorious markers such as *underscoring*, *comprehensively*, and *notably* are still rising. In other words, the AI-marker vocabulary is a moving target, and a block-list frozen in 2024 already misses the current generation. Any practical guideline therefore needs a regeneration procedure, not a fixed list. We also found that several folk markers fail empirically: *sheds light on*, *a plethora of*, and *myriad* did not increase after 2023 in this corpus. They are simply rare, in human and machine text alike.

3.5. Vanishing human phrases

Contamination is visible from the opposite direction as well. Several high-frequency human phrases are declining in the mixed-authorship years: *in other words* fell to 0.2 times its baseline rate, *on the other hand* to 0.3, and *it should be noted* to 0.4. A controlled experiment (Section 5) confirms that unguided LLM drafts contain essentially none of these expressions. This makes them attractive as positive guidance. Rather than only forbidding machine-flavored words, a guideline can ask the model to write with the connectives the field actually uses.

3.6. Venue conventions

The corpus also yields venue-specific norms that authors absorb by imitation. ICASSP papers prefer the heading CONCLUSION over CONCLUSIONS by a factor of 2.2, cite figures as “Fig. N,” and rarely number a discussion section. Interspeech papers show the reverse preference for CONCLUSIONS, cite “Figure N,” frequently include a numbered discussion section, and often number the acknowledgements. The dominant ICASSP skeleton is Introduction, Proposed Method, Experiments, Conclusion; the dominant Interspeech skeleton inserts Results and Discussion. Such details rarely appear in style guides, yet a draft that violates them reads as foreign to reviewers.

4. From measurements to guidelines

We compiled the results into a compact, machine-readable style reference. It contains the BAN list (31 words and 10 phrases with baselines below 3/M, each with its measured surge factor), the LIMIT list with explicit target rates (for example, *notably* at most once per paper, since its human rate of 24/M corresponds to 0.07 occurrences in a 2,700-word paper), the positive list of human markers and field formulas, the quantitative targets of Table 1, and the venue conventions of Section 3.5. Because inflected variants can evade phrase matching, the file also ships stem-based `grep` patterns for a final manuscript check. The reference is written as a skill file for LLM coding agents, so the same document serves both as documentation for humans and as an executable constraint for models. Regeneration is scripted end to end, from PDF collection to list synthesis, so the lists can be refreshed as marker fashion moves.

5. Validation

We tested whether the guidelines actually move generated text into the human range. For three published papers on turn-taking and backchannel modeling, we asked an LLM (Claude, Anthropic) to write an introduction on the same topic, once with no style instructions and once with the style reference. We then compared both drafts with the corresponding human introductions using the metrics of Section 2.3.

Table 4 summarizes the outcome. The unguided model already avoided the notorious BAN vocabulary, which is consistent with the fashion dynamics of Section 3.3, but

Table 4: Introduction-writing experiment (per 1k words; human markers as defined in Section 3.4).

Metric	Unguided	Guided	Human
Em-dashes	3.98	0.0	0.0
Human markers	0.0	6.88	6.05
BAN words	0.0	0.0	0.0
Mean sentence length	23–29	24.0	24.6

it opened its very first sentence with a paired em-dash insertion and used none of the human connectives. With the style reference, the em-dashes disappeared and the human-marker rate landed within 15% of the real papers. One BAN phrase slipped through as an inflected variant (“shed *some* light on”), which is precisely the failure mode the stem-based check now covers. The experiment is small, three topics and one model family, and the metrics are necessarily coarse. We regard it as a functional test of the guidelines rather than a claim about detection.

As a final check, we ran the compiled manuscript of this paper through the same measurement pipeline. The draft satisfied the em-dash, bullet, and rhetorical-question rules by construction. It did not satisfy the semicolon rule: the first version contained seven semicolons, a rate of 0.065 per sentence and roughly twice the human baseline of Table 1. Three of them were split into separate sentences, two sit in table captions, and the remaining two were kept deliberately. The version you are reading measures a human-marker rate of 5.5 per thousand words, within the range of the real introductions in Table 4. In other words, the guidelines caught their own authors, which is the behavior one would hope for.

6. Discussion

6.1. Style as a moving target

The peak-and-decline pattern of Section 3.4 suggests that scientific English is now coevolving with the models that help write it. Words become markers, markers become notorious, and notoriety drives suppression, both in model post-training and in author self-editing. In other words, the distribution we measured in 2026 will not be the distribution of 2028. This has two practical consequences. First, any static blocklist, including ours, has a shelf life of roughly one submission cycle, which is why the release ships the full regeneration pipeline rather than only its current output. Second, the more durable signal sits on the human side. The formulaic core of Table 2 has survived twenty-five years of paradigm shifts in the underlying technology, from GMM–HMM systems to end-to-end models, and it changed less in that time than the AI-marker vocabulary changed in two years. Guidance anchored to the stable human baseline should therefore age more gracefully than guidance anchored to the failures of any particular model generation.

6.2. Ethical considerations

Guidelines that make LLM-assisted text match human norms can be read as a tool for hiding LLM use, so the intended use deserves a clear statement. The guidelines encode the editorial norms of a research community, and following editorial norms is what careful human revision has always done. Our position is that assistance should be disclosed, as it is in this paper, and that the text itself should meet the standards of the venue regardless of how it was drafted. Note that the measurements cut both ways: the same tables that guide generation also sharpen the informal detector that readers already apply. What the community should avoid, in our view, is treating individual marker words as evidence of misconduct. Section 3.4 shows that some notorious words were never common in machine text for this field, and Section 3.2 shows that ordinary human words such as *crucial* are now statistically suspect. Word-level suspicion punishes authors, including non-native speakers who learned the suspect vocabulary legitimately, without establishing anything about authorship.

6.3. Limitations

The analysis inherits the limits of PDF text extraction, so paragraph boundaries and mathematical content are only partially recovered, and one venue-year (ICASSP 2025) showed en-dash and quotation-mark anomalies that we attribute to a typesetting change and excluded from the guidelines. Frequencies are corpus-level, so we cannot attribute any particular paper to LLM assistance, and our contamination estimates are lower bounds. The corpus covers two conferences and English text only, and the validation experiment used one model family and three topics. Finally, the guideline targets are frequencies, not meanings. A draft can satisfy every number in Table 1 and still be wrong, vague, or dull; the guidelines address style, and only style.

7. Conclusions

We measured 25 years of writing style in ICASSP and Interspeech proceedings and found a stable human baseline, an abrupt LLM-era discontinuity beginning in 2024, fashion dynamics that erode fixed blocklists, and a set of vanishing human phrases that provide durable positive guidance. The measurements were distilled into an evidence-based, regenerable style reference for AI-assisted writing, and a generation experiment showed that the reference moves LLM drafts to within the measured human range on the metrics studied. Future work includes subfield-specific phrase inventories, coverage of journals and of other research communities, and periodic regeneration to track the drift as it continues. This paper was drafted and revised under the proposed guidelines; whether it reads as human is left to the reviewer.

8. Acknowledgements

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